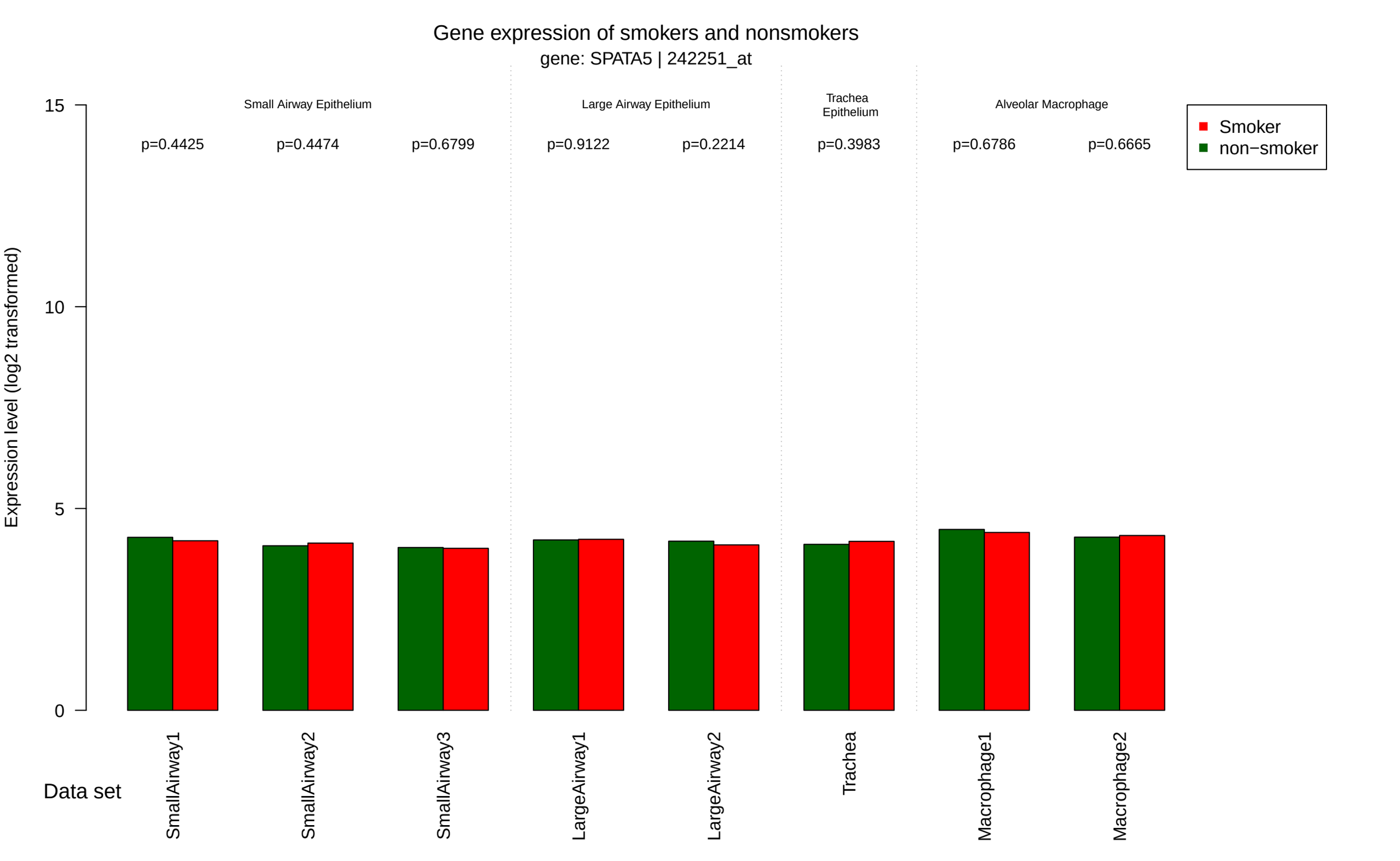
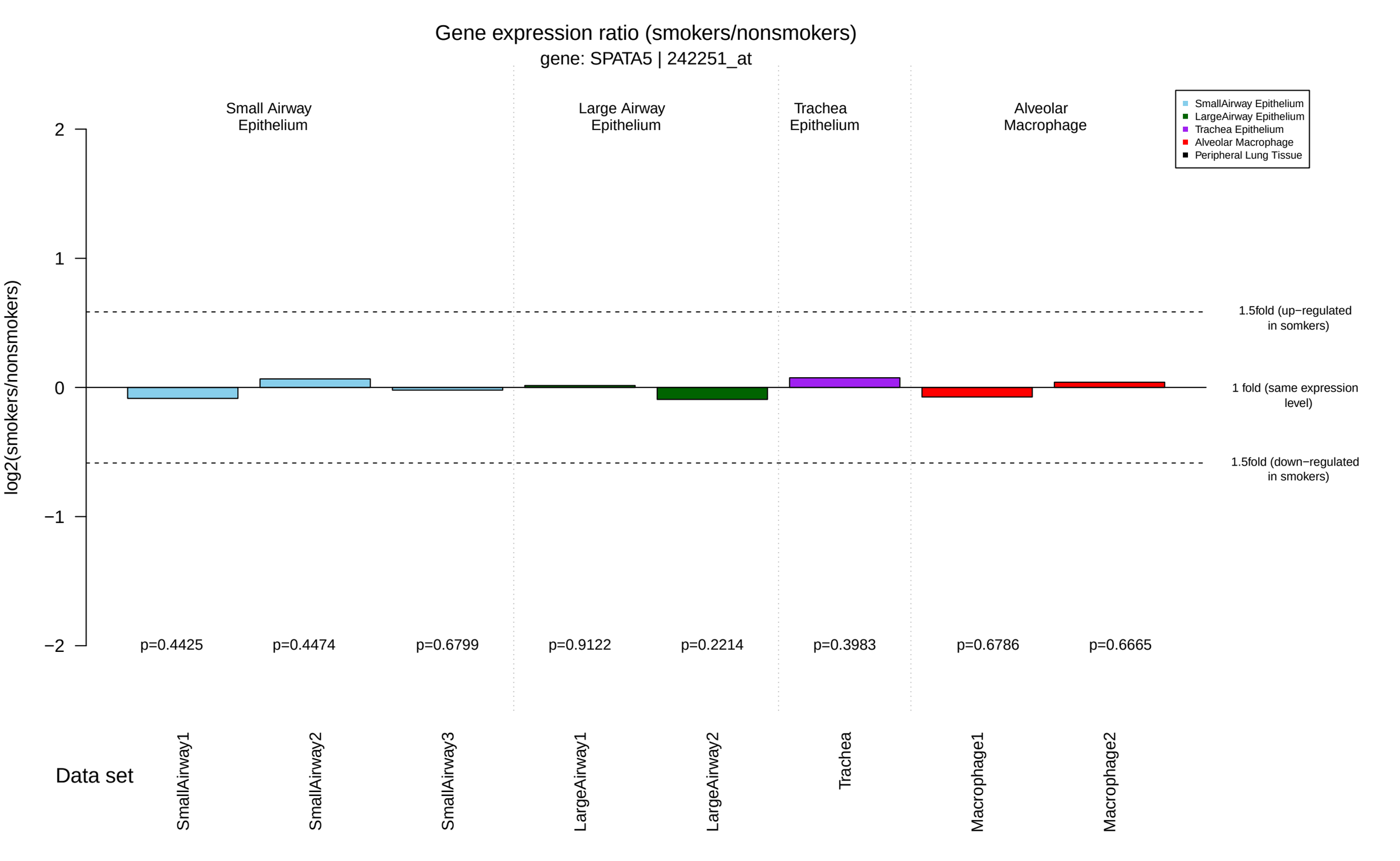
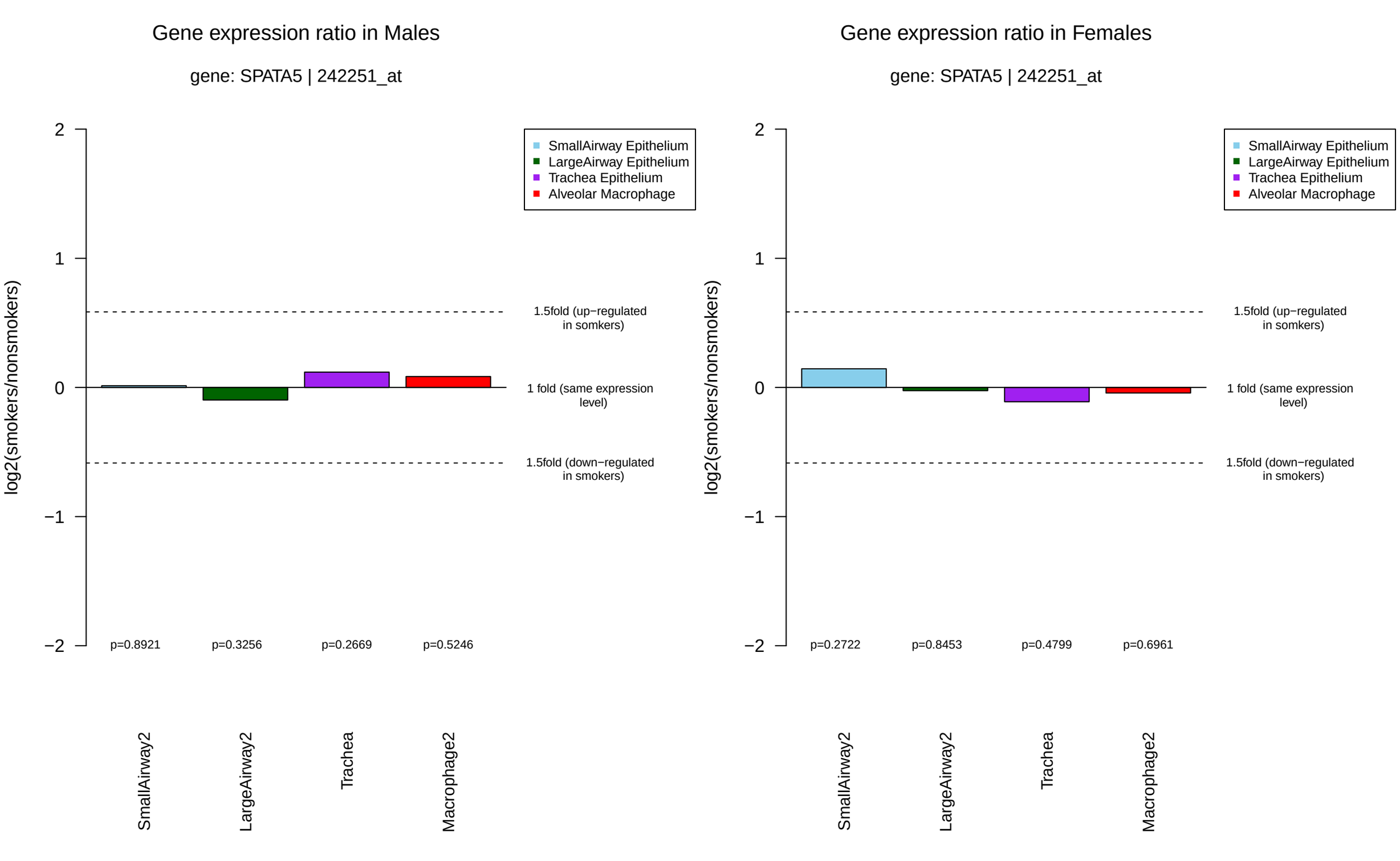
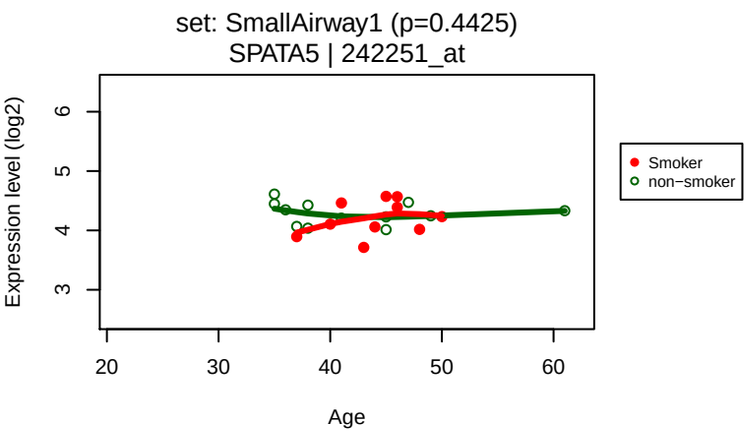
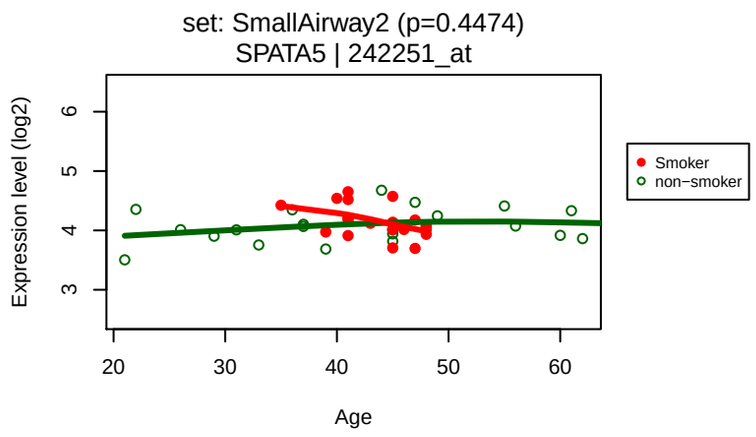
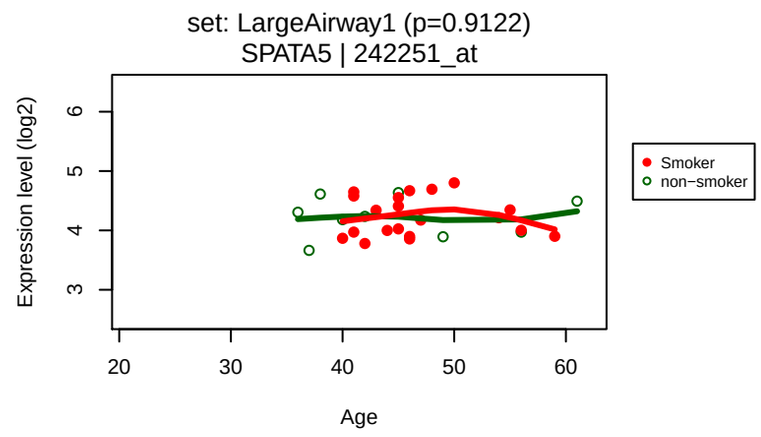

Gene expression of smokers and nonsmokers
gene: SPATA5 | 242251_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4425
p=0.4474
p=0.6799
p=0.9122
p=0.2214
p=0.3983
p=0.6786
p=0.6665
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

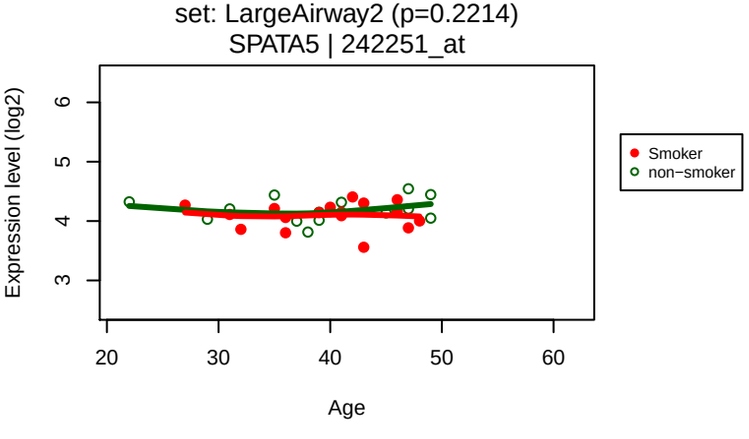

Gene expression ratio (smokers/nonsmokers)
gene: SPATA5 | 242251_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4425
p=0.4474
p=0.6799
p=0.9122
p=0.2214
p=0.3983
p=0.6786
p=0.6665
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

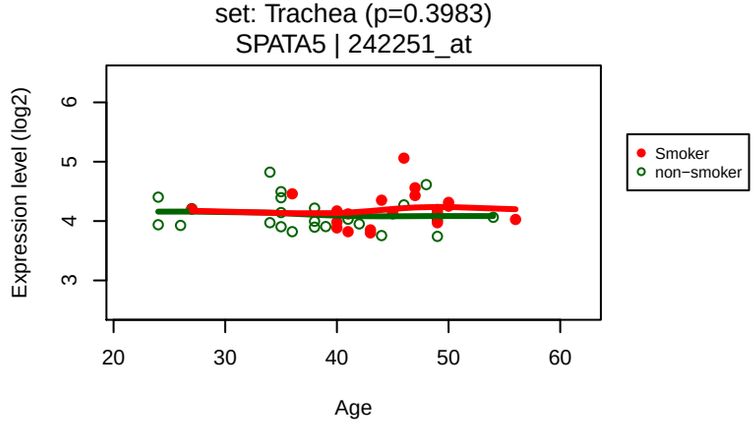

Gene expression ratio in Males
gene: SPATA5 | 242251_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8921
p=0.3256
p=0.2669
p=0.5246
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA5 | 242251_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2722
p=0.8453
p=0.4799
p=0.6961
SmallAirway2
LargeAirway2
Trachea
Macrophage2

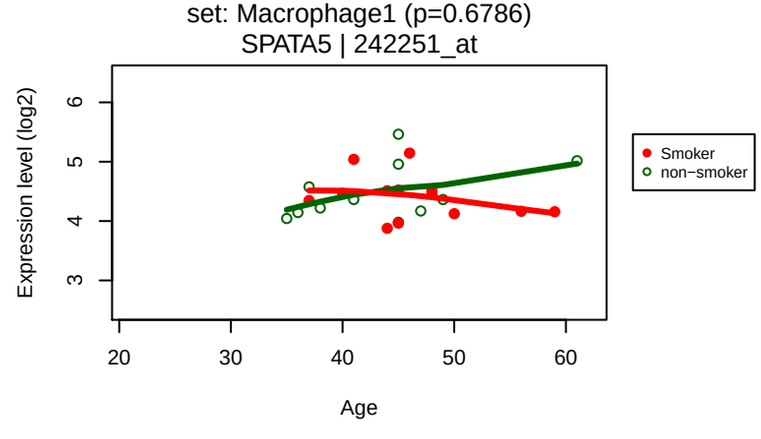

set: SmallAirway1 (p=0.4425)
SPATA5 | 242251_at
Expression level (log2)
Age
Smoker
non−smoker

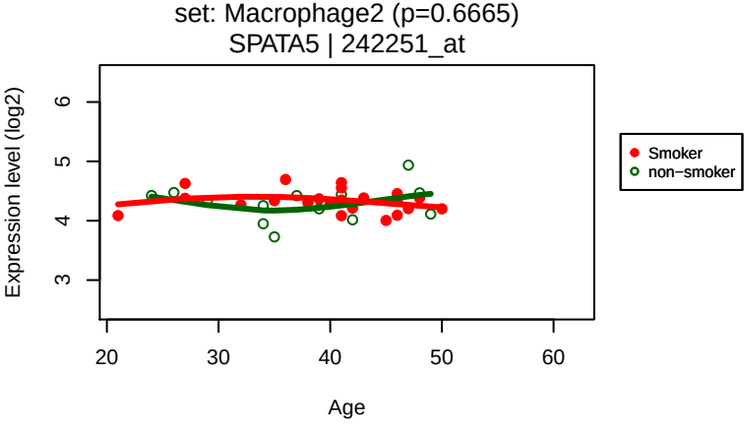

set: SmallAirway2 (p=0.4474)
SPATA5 | 242251_at
Expression level (log2)
Age
Smoker
non−smoker

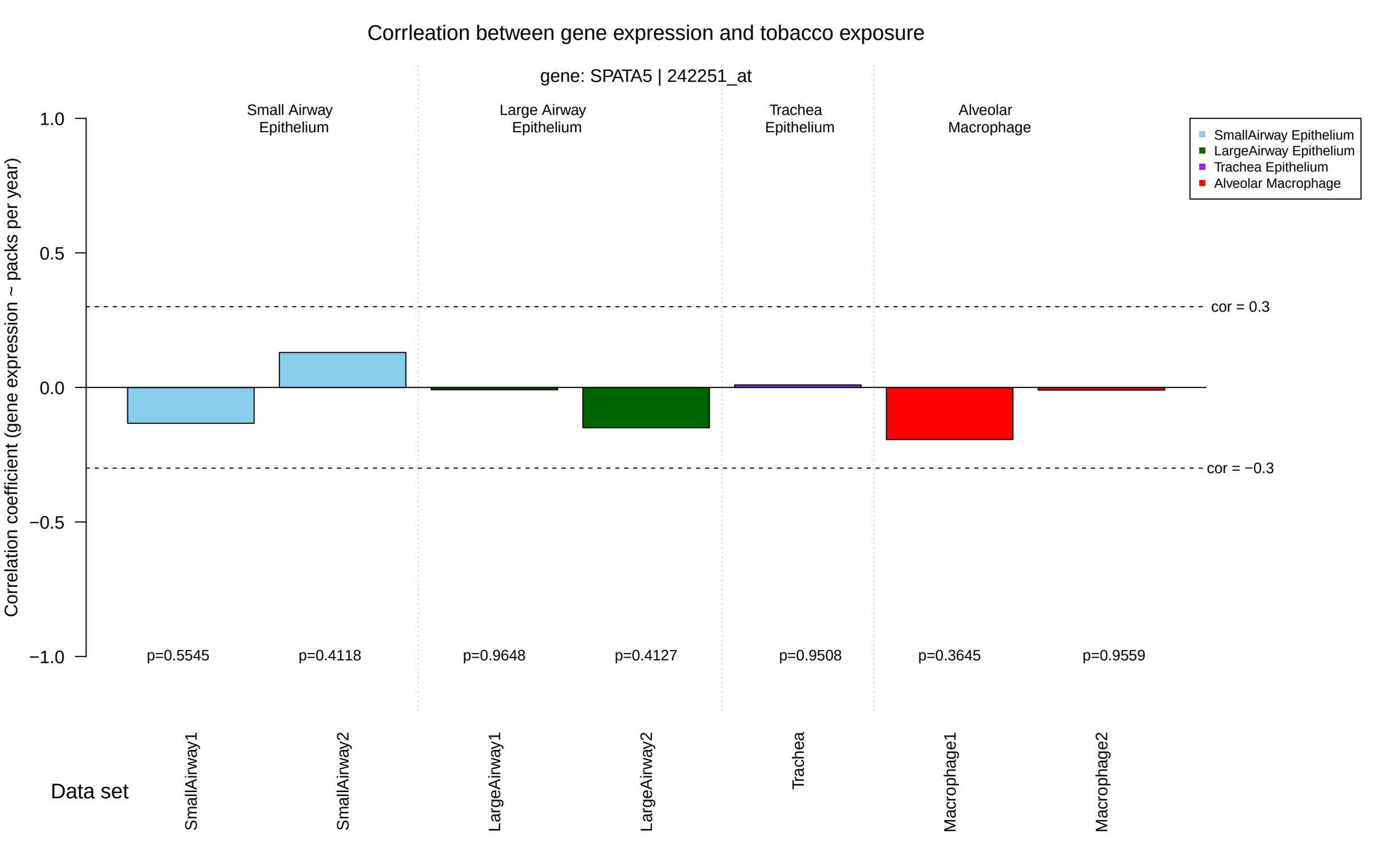

set: LargeAirway1 (p=0.9122)
SPATA5 | 242251_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2214)
SPATA5 | 242251_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3983)
SPATA5 | 242251_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6786)
SPATA5 | 242251_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6665)
SPATA5 | 242251_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA5 | 242251_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5545
p=0.4118
p=0.9648
p=0.4127
p=0.9508
p=0.3645
p=0.9559
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2